

3 (short)

(a) Derive an expression for the output voltage v_{out} in the circuit shown in Fig 3. The operational amplifier is ideal. [3]

(b) Consider two signal sources v_1 and v_2 with negligible output resistance. Design a circuit with two ideal operational amplifiers to produce a circuit output of $(200v_1 - 50v_2)$. You may decide the values of the resistances in your design. Justify your choice of resistance values. [7]

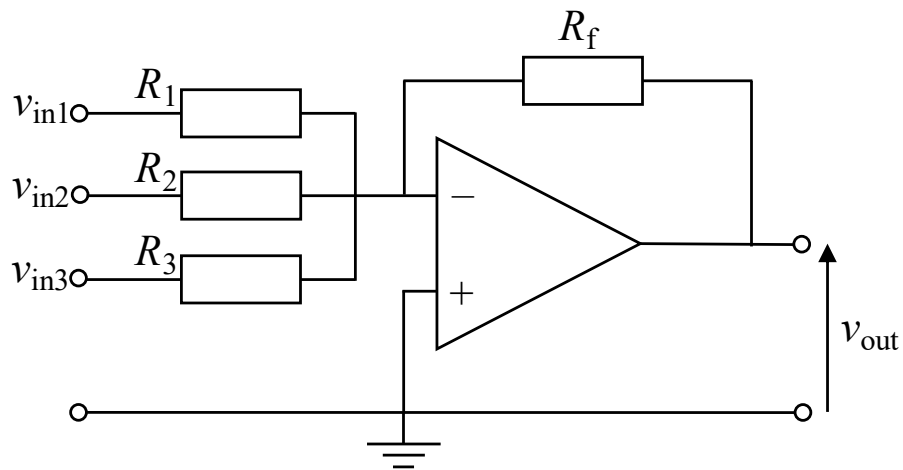


Fig. 3